

ENGINEERING WORKSHOP LAB SEMESTER PROJECT PROPOSAL

Title:

Automatic Water Dispenser

Member:

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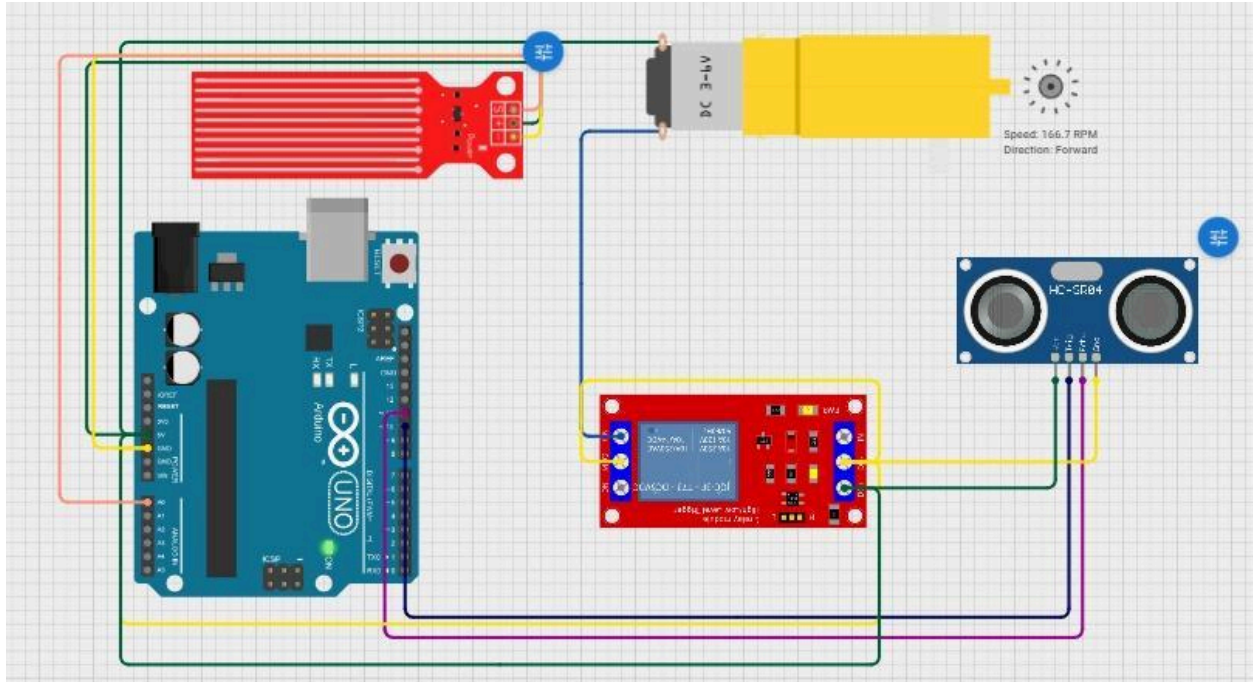
Problem Statement:

The Industrial Gap: In industrial environments, manual liquid distribution systems suffer from two critical failures: volumetric inconsistency and equipment damage due to dry-running. When operators manually fill containers, "over-filling" leads to chemical/resource waste, while "under-filling" compromises batch quality. Furthermore, if a supply reservoir empties without the operator noticing, the delivery pump can overheat and fail, leading to expensive downtime and maintenance costs.

Proposed Solution:

Smart Volumetric Dispensing with Asset Protection: The proposed system is an integrated Automated Fluid Management Unit. It utilizes an infrared/ultrasonic sensor for container detection to trigger a calibrated "Fixed-Volume" fill cycle, ensuring every glass/container receives the exact same amount of fluid regardless of human intervention. Simultaneously, a submersible water-level sensor monitors the reservoir. If the fluid level drops below a safety threshold, the system triggers a "Hardware Lockout" and an audible alarm, preventing pump damage and alerting the operator to refill the source.

Circuit diagram:



GANTT CHART:

